

Technology mediated routines and their effects on organizational performance: An integrated affordance theoretic approach

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ARTICLE INFO

Keywords:

Affordance theory
Organizational routine
Organizational change
Body-worn camera
Police organization

ABSTRACT

This study addresses a critical gap in Information Systems (IS) research that applies affordance theory. While previous IS affordance research has examined how technological affordance influences organizational routines or directly impacts performance, few studies have integrated these perspectives to explore how technology-induced routine changes mediate the relationship between technological affordances and organizational performance. To fill this gap, we first built an integrated affordance theoretic model. To validate this model, we collected survey data from 338 U.S. police departments focusing on using body-worn cameras (BWC) in their daily operation. Analyzing the survey data, we empirically validated that the technical affordances of BWCs (e.g., recording capabilities and infrastructural capabilities) drive changes in policing routines, which, in turn, enhance organizational performance. A primary contribution of this research is that we integrate qualitative and quantitative perspectives developed surrounding affordance theory in the IS research community, and validate the important role of routine changes in mediating the relationship between technological affordance and organizational performance. Our findings demonstrate that these technology-induced routine changes serve as key mediators, which exert a significant impact on performance beyond technology alone. This study underscores the importance of aligning technological affordance with organizational routines to drive sustainable improvements in organizational performance.

1. Introduction

As new technologies continue to sweep across industries, companies face pressure on how to adopt new technologies to derive business value from their technological investments. However, despite years of digitizing their operations, firms continue to struggle to implement technology successfully and to achieve higher levels of productivity. The Standish Group Chaos Reports on IT project failure rate spanning from 1994 to 2018 are alarming, with roughly 25 % of projects being outright failures, and more than a third being marginally successful, leaving only one-third of projects deemed as successful (Kappelman et al., 2022). Thus, how to turn IT investments into better performance both at the unit level and across entire organization remains an important IS research topic (Chae, Koh, & Park, 2018; Kappelman et al., 2022).

Information systems (IS) scholars have applied affordance theory to

examine this topic. Affordance is a relational concept considering that both technological and organizational factors are important to explain how technologies influence the direction of organizational development (Robey, Anderson, & Raymond, 2013; Volkoff & Strong, 2017). It has been widely employed to explore the processes that help organizations successfully implement new technology (Dremel, Herterich, Wulf, & Vom Brocke, 2020; Du, Pan, Leidner, & Ying, 2019; Tim, Hallikainen, Pan, & Tamm, 2020), the effective use of a specific technology by an organization (Robey et al., 2013; Volkoff & Strong, 2017), and the changes in organizational practices and structures accompanying technological adoption (Benitez, Braojos, Pavlou, & Llorens, 2023; Goh, Gao, & Agarwal, 2011; Hanelt, Busse, & Kolbe, 2017; Krancher, Luther, & Jost, 2018; Leidner, Gonzalez, & Koch, 2018; Leonardi, 2011; Boillat et al., 2015; Zhu and Jin, 2024). Indeed, affordance theory has significantly expanded our understanding of how organizations and

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<https://doi.org/10.1016/j.ijinfomgt.2025.102964>

Received 5 January 2025; Received in revised form 25 May 2025; Accepted 16 August 2025

Available online 29 August 2025

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technologies together shape organizational practices.

Qualitative research and quantitative research have shown different tendencies in employing affordance theory. While qualitative research has employed affordance theory to explain how technologies are implicated in the process of changing organizational routines, they have generally been indifferent to whether such changes lead to improvements in organizational performance (Boillat et al., 2015; Goh et al., 2011; Hanelt et al., 2017; Krancher et al., 2018; Leidner et al., 2018; Leonardi, 2011; Robey et al., 2013; Volkoff & Strong, 2017). In contrast, quantitative research has applied affordance theory to estimate the direct impact of technological affordances on performance outcomes in terms of innovation (Chatterjee, Moody, Lowry, Chakraborty, & Hardin, 2015; Chatterjee, Moody, Lowry, Chakraborty, & Hardin, 2020), knowledge sharing (Majchrzak, Wagner, & Yates, 2013; Pee, 2018), and cloud computing data security (Wang, Wang, Su, & Ge, 2020). However, they have overlooked the potential mediating role of technology-driven routine changes in the relationship between technological affordances and organizational performance. As a result, this gap in affordance theory has remained open between qualitative and quantitative IS research groups. We, therefore, propose two research questions:

RQ1: What are the effects of information technologies on changes in organizational routines?

RQ2: What are the effects of organizational routine changes on organizational performance?

To answer these questions, we propose an integrated affordance theoretic model, arguing that technological affordances can trigger changes in existing organizational routines, which, in turn, affect organizational performance. Then, we validate this research model with the case of the body-worn camera (BWC) technology used in U.S. police agencies. For this validation, we analyze survey data collected from 338 U.S. police agencies, and provide empirical evidence for the effects of information technologies on routine changes (i.e., RQ1), and the effect of routine changes on organizational performance (i.e., RQ2). Our analysis reveals statistically significant and positive results for both questions.

We add a theoretical contribution to the previous IS research employing affordance theory. We consolidate the strength of prior qualitative and quantitative research, which applies affordance theory. Prior qualitative research has focused on describing how the use of information technologies is implicated in the process of reconfiguring pre-existing organizational routines (e.g., use of technologies → organizational changes) (Boillat et al., 2015; Leonardi, 2011; Markus & Silver, 2008). However, this body of qualitative research has yet to measure the effect of routine changes on organizational performance. In contrast, past quantitative IS research has applied affordance theory to measure the direct effects of technological capabilities on performance in areas like innovation (Chatterjee et al., 2015; Chatterjee et al., 2020), knowledge sharing (Majchrzak et al., 2013; Pee, 2018) and data security (Wang et al., 2020) (*technology → performance*). However, these prior quantitative studies have been indifferent to the role of technology-induced *organizational routine*. Therefore, the presentation of our integrated affordance theoretic model (*technology → organizational routine → performance*) and its empirical validation make a significant theoretical contribution in addressing this gap. In addition, to the best of our knowledge, this study—drawing on a national survey of U.S. police departments—is one of the first to utilize the affordance perspective to explore the structural mechanisms through which BWCs impact organizational routines and performance within the context of law enforcement (Choi, Michalski, & Snyder, 2023; Gaub & White, 2020; Lum et al., 2020; Lum, Stoltz, Koper, & Scherer, 2019).

As a practical contribution, our findings offer valuable insights for law enforcement agencies considering the implementation of BWC technologies. Based on our findings, we highlight that the dynamic effects of BWCs on police departments range from micro-level changes in the routines of individual units to macro-level changes in overall organizational performance. This suggests that even incremental shifts at the

unit level can accumulate to produce substantial benefits for police agencies. This practical finding underscores the importance of aligning information technologies with the micro-routines of individual units, which can unlock performance improvements and help achieve broader organizational goals.

The article is structured as follows. We first discuss theories on organizational change (i.e., routine theory and affordance theory) to set the stage for analyzing the case of BWC technologies in U.S. police departments. We then develop the theoretical foundations for our hypotheses. Next, we discuss the design of the study, the data used in our hypotheses testing, and our empirical findings. We highlight the paper's contributions to theory and practice, examine its limitations, and conclude with closing remarks.

2. Theories of organizational change

2.1. Routine theory: from human-centric to materialist

Organizational scholars define an organizational routine as “a repetitive, recognizable pattern of interdependent actions, involving multiple actors” (Feldman & Pentland, 2003, p. 96). As organizational blueprints, routines serve as repositories of organizational structure, habits, and memories that provide employees with a cognitive framework for appropriate behaviors and performance targets (Nelson & Winter, 1985). While routines offer a stability and consistency for daily interdependent actions, organizations often deliberately modify them to implement strategic initiatives (Glaser, 2017; Rerup & Feldman, 2011), integrate new technologies (Barley, 1986; Edmondson, Bohmer, & Pisano, 2001), adapt to environmental changes (Kaplan, 2015; Zbaracki & Bergen, 2010), or enhance overall performance (Becker, Lazaric, Nelson, & Winter, 2005; Bresman, 2013; Knott, 2003). For this reason, organizational routines are often understood as a micro-foundation shaping different dynamic capabilities of an organization (Felin, Foss, Heimeriks, & Madsen, 2012) that enable organizations to continuously and stably evolve and avoid stagnation.

To articulate the repetitive and changing nature of routines, IS and management scholars have explored the dynamic interplay between ostensive and performative aspects of organizational routines (D'Adderio, 2008; D'Adderio, 2014; Dionysiou & Tsoukas, 2013; Feldman & Pentland, 2003; Pentland & Feldman, 2008; Pentland, Feldman, Becker, & Liu, 2012; Rerup & Feldman, 2011). The ostensive aspect of routines serves as a guiding template for what actions ought to be taken by all organizational members when performing their everyday tasks. The performative aspect of routine refers to actions that individuals take to apply the ostensive aspect of organizational routine in particular situations. Taken together, the ostensive aspect of routine provides a template for actions, while the performative aspect allows room for individuals in particular situations to deviate from those prescribed templates to produce better organizational outcomes (Becker et al., 2005; Glaser, 2017).

The interplay between the ostensive and performative aspects of organizational routines supports organizational stability while enabling dynamic adaptation to changing business environments. For instance, changes in the ostensive aspect frequently trigger corresponding adjustments in the performative aspect, and ensure that new routines are actually enacted. Conversely, the performative aspect can also trigger changes in the ostensive aspect when employees in a particular situation intentionally deviate from the prescribed routines to yield better outcomes (Howard-Grenville, 2005). Especially, when such deviations prove effective, they can prompt broader changes in multiple actors across the organization, and lead to updates in the generalized template (i.e., the ostensive aspect) that guides organizational actions. Regardless of whether the change originates in the ostensive or performative aspect, the result is an update to the existing organizational routines accompanied by improvements (e.g., increased accuracy, increased efficiency, reduced time or effort) (Becker, 2005; Pentland et al., 2012).

Early routine theory has been framed as human-centric, envisioning organizational changes as primarily driven by individual employees who carry out the performative aspect of routines (Leonardi, 2011). Subsequent research has introduced materialist routine theory, which argues that the agential capabilities of material artifacts (i.e., non-human information technologies) also entail performative aspects in that they can trigger change in organizational routines (D'Adderio, 2011; Leonardi, 2011; Murray, Rhymer, & Sirmon, 2021; Pentland, Hærem, & Hillison, 2011; Robey, Anderson, & Raymond, 2013). For example, Pentland et al. (2011) analysis of invoice processing routines shows that 65 % of the required actions were carried out by people, while the remainder were performed by an automated workflow system. The usefulness of materialist routine theory becomes evident when considering how information technologies and broader infrastructure systems (e.g., power grids, traffic control systems, e-banking, e-shopping, e-government systems, security trading, supply chain, etc.) support the everyday routines of modern businesses (Morgan, 2002). The core message of materialist routine theory is that technological artifacts are so ingrained in organizational processes that it is nearly impossible to explain organizational routine changes without considering the technological capabilities used in organizations (Robey et al., 2013). Affordance theory, which recognizes both human and technological capabilities, has thus expanded our understanding of how organizational routines experience change.

2.2. Affordance theory as a materialist routine theory

Affordance theory contributes an additional perspective for understanding the interplay between technologies, human users, and organizational routine changes. Originating in ecological psychology (Gibson, 1979; Gibson, 1977), the concept of affordance considers how surrounding environments shape conditions for living beings to perceive, act, and evolve. This relational idea has been further refined by ecological psychologists (Chemero, 2003; Hutchby, 2001; Stoffregen, 2003) who examined how the capabilities of living organisms continuously evolve by composing, decomposing, and refining their relationship with external material environments. Within this framework, the performative capabilities of organisms cannot be reduced solely to their intrinsic attributes of living beings (i.e., subject) or to the material properties of their environment (i.e., object). Instead, affordances emerge through the recurring patterns of action that align with the constraints and possibilities offered by a given environment (Hutchby, 2001).

Importantly, the potential actions afforded to individuals by their environment are not limitless or random. Certain patterns of action become possible through repetitive and habitual engagements with the external environment, while others are restricted due to structural, material, or perceptual constraints (Leonardi, 2011). These varying action possibilities may arise from the physical limitations of the environment (Gibson, 1979), technological design choices (Norman, 1999), institutional norms (Leonardi, 2011), or cognitive biases that shape how individuals interpret their surroundings (Hutchby, 2001). For example, when the goal is privacy or protection, an actor may perceive a locking mechanism on a door as enabling as it can afford the possibility of barricading. In contrast, the same actor may experience the mechanism as constraining if the locking mechanism prevents them from escaping a threatening situation (Volkoff & Strong, 2017). This example demonstrates that affordances are relational in that, depending on the actor's goal, they may experience the same locking mechanism as either enabling or constraining.

IS research has embraced affordance theory to help explain technology-induced routine changes in organizations (Markus & Silver, 2008; Volkoff & Strong, 2017). Leonardi (2011) explains why incumbent work routines change when an organization introduces new technologies. He put, if an organization realizes a new technology as a constraint in its existing work routines, it tends to make alterations to

the technology so that certain technological features fit with the existing routines and help achieve organizational goals. By contrast, if an organization sees that a new technology provides new capabilities to achieve its goals, it tends to alter its work routines to accommodate the new technological capabilities to better attain those goals (Leonardi, 2011). In affordance theory, therefore, technologies are not viewed as inert objects which function as a component part of an organizational machine. Instead, they are understood as performative agents that can empower or limit organizational behaviors to certain directions. Leonardi (2011) uses the term *imbrication* to explain the way that organizational routines evolve as a result of interactions between the agential capabilities of technologies and human users. Its implication is that the repetitive use of certain technological features in particular organizational environments brings various changes to work patterns that eventually lead to the establishment of new organizational routines (Leonardi, 2011; Robey et al., 2013).

Acknowledging the performative aspects of both humans and technology in shaping routines, IS researchers have produced a plethora of qualitative studies describing how the selective and repetitive use of information technologies bring about changes in organizational routines, structures, and practices (Volkoff & Strong, 2017). These studies have advanced our understanding of how organizations adopt new technology (Du et al., 2019; Tim et al., 2020), effectively use new technology (Robey et al., 2013; Volkoff & Strong, 2017), or experience technology-introduced changes in organizational routines and practices (Boillat et al., 2015; Goh et al., 2011; Hanelt et al., 2017; Krancher et al., 2018; Leidner et al., 2018; Leonardi, 2011).

While these qualitative studies have helped explain how organizations evolve in tandem with the use of technologies, they have not been extended to examine whether technology-induced routine changes might lead to changes in organizational performance. Quantitative IS research, in contrast, has employed affordance theory to measure the direct effects of technological affordances on performance in such areas as innovation (Chatterjee, Moody, Lowry, Chakraborty, & Hardin, 2015, 2020), knowledge sharing (Majchrzak et al., 2013; Pee, 2018), cloud computing data security (Wang et al., 2020), new product development (Benitez, Braojos, Pavlou, & Llorens, 2023), and competitive advantages (Zhu & Jin, 2024). These quantitative studies describe how information technology's capacities can help in improving important performances valued by the organization. However, they have not considered the potential mediating effect of technology-afforded routine changes in the relationship between technological affordances and organizational-wide performance. Our study addresses this research gap through the case of BWC implementations in U.S. police agencies. Integrating prior qualitative and quantitative IS research that uses affordance theory, we examine if the performance of police organizations is impacted by organizational routine changes shaped by using BWC technology.

2.3. Literature review on BWC research

BWC systems represent an interesting case for studying the relationship between technological affordances, routine changes, and organizational performance. Since 2010, police agencies across the U.S. have increasingly adopted BWCs in response to public demands for accountability, particularly following several high-profile civilian killings by police (Fan, 2018). By capturing police-citizen interactions into video footage, the implementation of BWCs in police agencies has aimed to reduce police officers' use of force and increase police transparency, accountability, and public trust (Adams & Mastracci, 2019; Crow, Snyder, Crichlow, & Smykla, 2017; Cubitt, Lesic, Myers, & Corry, 2017; Gaub, Choate, Todak, Katz, & White, 2016; Hedberg, Katz, & Choate, 2017; Maskaly, Donner, Jennings, Ariel, & Sutherland, 2017).

However, this extensive research has shown that the effects of the use of BWCs in police agencies are inconsistent (Lum et al., 2020; Maskaly et al., 2017). While many studies have found that the use of BWCs can reduce the use of force by police officers, citizen resistance to police

officers, and civilian complaints (Gaub & White, 2020; Lum et al., 2019), other research has found no impact on the similar measures (Gaub & White, 2020; Grossmith et al., 2015; Hedberg et al., 2017). Moreover, some studies even have shown that wearing BWCs may increase assaults on officers (Ariel et al., 2016, 2018; Whynot, Nykorchuk, Zisis, & Deane, 2016). In addition, while BWCs have been found to reduce the number of citizen complaints against police officers, it remains unclear whether these reductions are due to an improved quality of police–citizen interactions or a change in reporting procedures (Lum et al., 2020).

Research on the impact of BWCs on police agencies at the organizational level is scant (Lum et al., 2019). Nonetheless, existing work provides important paths for considering the interplay of technological affordances, organizational routines and performance. Koen, Willis, and Mastrofski (2019), for instance, found that BWCs enhance police agencies' routines that have "clear goals and clear means for their accomplishment" (e.g., report writing, resolution of civilian complaints) (p. 13), but less so when goals and means are less technically defined or specified (e.g., supervision and training). Guzik, Sesay, Oh, Ramirez, and Tong (2021) report that the materiality of BWCs "affords" officers and agencies the ability to modify their work routines, including how officers carry out patrol work, how departments process requests for information from prosecutors and the public, and how departments resolve citizen complaints.

Prior research on BWCs highlights two key technological affordances of BWCs: the mobile-recording affordance (MRA) and the general IT affordance (GIA) (Guzik et al., 2021; Koen et al., 2019; Lum et al., 2019; Miranda, 2022). The MRA refers to the technological capability of BWCs for police agencies to capture and record their interactions with citizens while on duty. Unlike earlier recording devices, such as dash cameras or surveillance cameras, the MRA of BWCs enables police officers to move freely while capturing video and sound footage of policing activities from their first-person perspective. Therefore, the MRA provides a more comprehensive, accurate, and situational record of incidents than fixed video recording devices.

In addition, the GIA of BWCs provides capabilities to flexibly access, retrieve, and share stored video footage through a central data storage system. Police departments collaborate with BWC vendors to establish supporting data infrastructure and reconfigure their existing IT infrastructure as necessary. This integration ensures that the recording capabilities of BWCs function in harmony with their broader IT infrastructure, a concept we refer to as the GIA. In line with the main ideas of affordance theory, we emphasize that the reconfiguration of the GIA can initiate changes in pre-existing organizational routines. In the context of our research, the impact of GIA is particularly evident in the way police agencies manage digital evidence and conduct police operations. For example, GIA-initiated routine changes include the ways police agencies share BWC footage with the public in response to citizen complaints, transmit digital evidence to prosecutors' offices, and streamline how police agencies access and review footage for everyday operations (Guzik et al., 2021; Koen et al., 2019).

To date, no studies have examined the intersecting dynamics of technological affordance, organizational routines, and performance outcomes in their entirety. Thus, we use an exploratory quantitative approach to assess the impact of two BWC affordances (i.e., MRA and GIA) on organizational routines, as well as their subsequent impact on performance in police agencies.

Among the various routines in police agencies, we focus on the case reporting routine and the officer supervision routine, which have been identified as the most likely to undergo changes resulting from repeated uses of BWCs (Guzik et al., 2021; Koen et al., 2019). In practice, these are important routines in police agencies (De Keijser, Malsch, Kranendonk, & De Gruijter, 2012; Kahn & Martin, 2016). Case reports, which provide detailed accounts of focal events and the evidence-gathering process, serve as the main source of information for court proceedings (Boivin & Gendron, 2021). Therefore, producing accurate and comprehensive case

reports is important to facilitate fair and transparent investigations, prosecutions, and adjudications. Officer supervision is another important routine, as it involves review and assessment of the job performance of police officers, which should adhere to professional and ethical work protocols. Provided that the deployment of BWCs in police agencies aims to promote transparency and accountability for improved law enforcement activities, it is reasonable to examine how these two routines may undergo changes through everyday use of BWCs (Guzik, Sesay, Oh, Ramirez, & Tong, 2021; Koen, Willis, & Mastrofski, 2019; Pelfrey & Keener, 2018; Vredeveltdt, Kesteloo, & Hildebrandt, 2021).

As dependent variables, we focus on two performance measures in police departments: policing performance and public relationship performance. In the field of criminology, policing performance is largely defined by crime rates, clearance rates, response times, and enforcement activities such as arrests, citations, and so on (Rosenbaum et al., 2017). These are typical measures of police performance that has been frequently used in previous BWC research (Grossmith et al., 2015; Henstock & Ariel, 2017; Hughes, Campbell, & Schaefer, 2020; Pyo, 2021). On the other hand, public relationship performance is another important measure that estimates the effectiveness of police agencies based on the trust and legitimacy they gain from the communities they serve, which, in turn, enhances support for their public safety efforts (Rosenbaum et al., 2017).

3. Hypotheses development

3.1. Impacts of MRA on case reporting and officer supervision routine

Previous research on affordance theory argues that new technology offers the potential for organizations to initiate certain actions to achieve their goals (Duan, Deng, & Wibowo, 2024; Leonardi, 2011; Robey et al., 2013; Volkoff & Strong, 2017). In the context of police agencies, the MRA of BWCs alters case reporting routines by providing police officers with visual and auditory records of their interactions with focal incidents. The richness of information recorded in BWC footage serves as a useful input for improving the detail and accuracy of case reports (Guzik et al., 2021; Koen & Willis, 2020; Koen et al., 2019). It can also assist officers in recalling important details that might otherwise be forgotten while conducting policing duties under high-pressure and complex situations that demand heightened situational awareness. Thus, BWC footage provided by the MRA can lead to changes in how police officers write case reports. Before using BWC footage, police officers relied solely on their field notes and memories to write case reports (Vredeveltdt et al., 2021). However, as they can now also refer to BWC footage, they can improve the reliability and credibility of their reports by adding footage identifiers to the case reports (Newell & Koen, 2023; Vakhitova, Iliadis, Harris, Tyson, & Flynn, 2023; Vredeveltdt et al., 2021). Thus, the MRA can change the case reporting routine (H1a).

In addition, the rich information captured in video footage can transform the routine of officer supervision (H1b) (Guzik et al., 2021; Koen et al., 2019). Because BWC footage provides visual and audio records of interactions between police and citizens, supervisors can review police–citizen interactions more easily and in greater detail than before (Campeau & Keesman, 2024). Also, by reviewing recorded interactions, supervisors can better evaluate if officers' interaction with citizens comply with regulatory standards and provide personalized feedback to officers (Willis, 2022). Therefore, it is reasonable to anticipate that the routine of officer supervision will evolve as police agencies increasingly and repeatedly use BWCs (H1b). Therefore, we hypothesize that:

H1. : MRA has positive impacts on the change in the routines of (a) case reporting and (b) officer supervision within police agencies.

3.2. Impacts of GIA on case reporting and officer supervision routine

The GIA, when configured to work with BWCs, can trigger changes in

case reporting and officer supervision routines (H2). GIA refers to the capability of an extended IT infrastructure to manage recorded video footage through saving, locating, retrieving, and sharing. These extended capabilities can significantly impact these routines by allowing officers to efficiently access and retrieve video footage (Boivin & Gendron, 2021; Guzik et al., 2021; Koen et al., 2019) upon request. For example, police officers can now locate and retrieve recorded videos from a storage system – whether in-house or cloud-based – through software, apps, or web platforms using case numbers. This ease of access enables officers to complete case reports more accurately and efficiently. One interviewee¹ shared with us:

“if you have a domestic violence or something, and it’s something physical, and you got to get in there and take care of it. But then if you go back and be like, ‘I didn’t know what that person’s name was’, you could go back and pull it up. ‘Oh, it was Tom so and so, or whatever’. That information can help make your reports more accurate.”

The above interview suggests that the GIA enables police officers to access and review BWC footage when preparing case reports. These technical features, which allow officers to access, locate, and review recorded video footage, greatly aid in writing accurate and comprehensive case reports by supplementing their porous memory and field notes. This capability is especially valuable for police officers, as it enables them to focus on their on-site policing duties, which often escalate into violent situations and impair their ability to recall their interaction with precision (Yu & Monas, 2020). On the other hands, an empirical study reports that, when writing reports, officers are more likely to review footage, if they can more easily locate a particular video (Koen, Newell, & Roberts, 2021). Therefore, as officers increasingly use the GIA-supported video footage, it is likely to transform their case reporting routines over time (H2a).

In addition, the GIA is also likely to influence the supervision routine of police departments. Internal affairs officers and other supervisors now rely less on traditional interviews with officers involved in civilian complaints to assess the appropriateness of police interactions. One police officer that we interviewed noted:

“We’d interview her [the complainant], we’d interview the officer, and a lot of times that would come out as not sustained if we didn’t actually have proof from either one... With a body worn camera, we can review it, and, actually we’ve invited people to come in to review the video. Because if it’s something, so, let’s say, again, it’s the perception of the person complaining, and I’m looking at the video and it’s completely opposite of what that person is saying, I definitely would invite them in to come and take a look at it... They’re like, ‘Oh, forget it, and then that’s the end of it’.”

The interview illustrates how the GIA, which supports easy access to and review of video footage through its cloud-based storage services, is likely to transform the supervision routine. Specifically, it highlights a GIA-driven change from reliance on traditional subjective interviews to more objective and evidence-based oversights of police interactions with citizens. This example underscores the role of the GIA in enabling easy access, location, and replay of video footage, empowering supervisors to quickly verify claims by inviting both officers and complainants to review the footage for transparent resolution of discrepant claims.

In summary, the GIA, which simplifies the process of locating, accessing and reviewing video footage and other evidence files (e.g.,

pictures, notes, case reports etc.) with corresponding case numbers, is likely to introduce changes to the routines of case reporting and officer supervision. By streamlining these tasks and reducing the time needed to access and review digital evidence, the GIA is likely to encourage police officers to integrate its technical capabilities into their workflow more frequently. This repeated reliance on the GIA is likely to gradually reshape how routine tasks are performed. Therefore, we propose the following hypothesis:

H2. : GIA has positive impacts on changes in the routines of (a) case reporting and (b) officer supervision.

3.3. Impacts of changes in the case reporting routine on performance

Organizational routines have been found to be fundamental for performing daily activities and they do not operate in isolation (Geiger, Danner-Schröder, & Kremser, 2021). Thus, when a routine changes, it can generate broader organizational impacts by carrying over the changed effects to other dependent routines (Pentland, Recker, & Wyner, 2016). Likewise, changes introduced by BWCs in case reporting routines can generate broader effects in police departments by providing improved quality documentation for police operations.

The BWC-enabled changes in case reporting routines can influence police department performance in two important ways. First, improved documentation practices supported by BWC footage enable the creation of more accurate and comprehensive case reports (Boivin & Gendron, 2021; Yu & Monas, 2020). These high-quality case reports serve as critical inputs for core dimensions of policing performance (PP), including the efficient resolution of citizen complaints, the development of evidence-informed strategies to reduce crime rates and police use of force, and the provision of reliable evidence to support legal proceedings (Guzik et al., 2021; Koen et al., 2019)

Second, improvements in case reporting routines are likely to enhance the on-the-ground policing performance of frontline officers by reducing the cognitive and physical burdens associated with their field duty. Traditionally, officers had to engage in complex real-time interactions with citizens, while also observing, memorizing, and organizing key details for later report writing. This kind of multitasking under high-pressure conditions often strained officers’ situational awareness and compromised the policing performance (Baldwin et al., 2022). In this regard, the BWC-enabled case reporting routines alleviate this multitasking pressure by allowing officers to rely on recorded footage when writing case reports back at the department, rather than having to take notes and mentally record various events and situations during field interactions. This new mode of case reporting routine re-locates the documentation tasks from the field to a safer and controlled setting so that officers can focus more on the field interaction itself, such as reading body language or and detecting behavioral cues. Detective Hawthorn of the Kentridge Police Department articulated this changed dynamics as follows

“[before using BWC systems] I was so busy writing notes, I couldn’t focus my attention on the body language, on the changes in an individual. It was harder for me to detect the deception or what I perceived to be those clues that we were trained in. I can focus more now in-person”.

What this account illustrates is that the new case reporting routine, enabled by using BWC systems, contributes to enhancing officers’ policing performance by helping them assess situation more accurately and respond more effectively. Therefore, the changes of case reporting routines are likely to provide officers with greater capacity to manage incidents, enhance community safety, and document operationally relevant evidence, ultimately contributing to improved policing performance (H3a).

The BWC-enabled changes in case reporting routines are also likely to improve the public relationship performance (H3b). By supporting

¹ We conducted 10 semi-structured interviews at the Kentridge police department to capture an initial understanding of how police agencies use and experience BWCs. To preserve their privacy, we use pseudonyms for the department and the police officers who participated in our study. Please refer to Appendix A for a more detailed description of the semi-structured interviews conducted as part of this exploratory study.

the production of more accurate, detailed, and comprehensive case reports, these improved case reporting routines are likely to enhance the police department's capacity to respond transparently and reliably to inquiries from the public or media. This enhanced capability not only increases institutional transparency and accountability, but also helps foster greater public trust in law enforcement (Yu & Monas, 2020). Moreover, well-documented case reports created through BWC-supported case reporting routines can play a key role in addressing public concerns, correcting misinformation, and facilitating clearer communication between police departments and the communities they serve. In doing so, BWC-enabled case reporting routines can directly support the goal of improving public relationship performance (White, Todak, & Gaub, 2018). Therefore, we propose the following hypothesis:

H3. Changes in case reporting routine have a positive impact on (a) police performance and (b) public relationship performance.

3.4. Impacts of changes in officer supervision routine on performance

Changes in officer supervision routines, afforded by the routine use of BWC footage, are likely to improve policing performance (H4a). Integrating BWC footage into everyday supervisory practices provides supervisors with more objective, detailed, and situationally relevant material to assess officer behaviors (Campeau & Keesman, 2024). One of its key benefits is the capability for supervisors to review BWC footage collaboratively with individual officers or teams to assess whether their conduct aligns with departmental protocols and situational best practices. This approach enables supervisors to deliver more targeted, evidence-based feedback tailored to specific actions and decisions (Owens, Weisburd, Amendola, & Alpert, 2018). Over time, such routine supervision can promote consistent adherence to best practices in citizen interactions, reduce use of force by officers, and address citizen complaints etc. In doing so, BWC-supported supervision routines not only strengthen ongoing officer development and morale but also help align day-to-day officer behavior with the broader policing performance goals of police departments (H4a).

Whereas the benefits in H4a concern internal capacity-building of police departments, H4b emphasizes how improved changes in supervision routines can be perceived externally by the community. Changes in officer supervision routines are likely to generate a positive impact on public relationship performance (H4b). The use of objective BWC footage in supervision reinforces standards of professionalism and ethical conduct, leading to more consistent and fair policing practices (Engel, 2002). When supervisors rely on BWC-generated evidences to routinely monitor police conduct and address performance issues proactively, they signal a visible commitment to fairness and integrity (Demir, 2023). This accountability mechanism can not only improve public confidence in how officers are trained and held to standards, but also help strengthen the legitimacy of law enforcement in the eyes of the community (Cohen, McCormick, & Rich, 2019). By making supervisory practices more transparent and ethically grounded, police departments can foster stronger relationships with the community and build trust over time (Moura, Borges, Morgado, & Ramalho, 2023). In summary, while changes in supervision routines improve police performance by refining officer behavior (H4a), they also improve public relationship performance by enhancing perceptions of legitimacy and accountability (H4b). Therefore, we propose the following hypothesis:

H4. Changes in officer supervision routine have a positive impact on (a) police performance and (b) public relationship performance.

4. Research design and data analysis

4.1. Survey administration and data collection

To empirically test our hypotheses, we conducted a nationwide survey targeting U.S. police departments that had adopted. We

contacted police agencies in municipalities ranking in the top 20 % of the U.S. based on population. These included 5214 police agencies across all 50 US states, ensuring a focus on departments with substantial operational capacity and diversity. The survey was developed using the Qualtrics platform. To analyze the collected data, we employed partial least squares structural equation modeling (PLS-SEM) using ADANCO software.

From June to December 2021, we emailed our survey links to the 5214 police departments and sent three reminders encouraging them to complete the survey. Ultimately, 1045 responses were received. Of these, 157 departments indicated that they do not use BWCs, resulting in 888 responses. We removed 463 responses having more than 3 unanswered items, 10 responses missing reversed scales, and 1 response having the same answers for all items, leaving 414 completed surveys. To minimize bias resulting from disruptions to departmental operations during the Covid-19 pandemic, we excluded 76 responses from police agencies implementing BWCs from 2020 to 2021. As a result, 338 responses were used for the analyses.

4.2. Measures

We combined existing scales with constructs developed from our semi-structured interviews with 10 police officers (refer to footnote 1 in page 16 for details). The scales for the dependent variables (i.e., policing performance and public relation performance) were adopted from previous BWC literature (Smykla, Crow, Crichlow, & Snyder, 2016; Tankabe & Ariel, 2016) and tailored to our research. Given the limited availability of validated measures for BWC affordances and routine changes, we developed new items to capture these constructs.

Our scale development followed a multi-step process. First, we compiled and analyzed a comprehensive list of BWC technical specifications (Reinhardt, 2015) and routine characteristics (Becker, 2005; Becker et al., 2005). This review ensured that our items reflect the capabilities of BWCs and changes in case reporting and office supervision routines. Second, we drafted the initial items based on the definition of the constructs, and formed a panel of experts from sociology, organizational science, and information systems to review these items for face validity. Each member independently assessed whether these items captured each aspect of each construct domain. Their feedback was instrumental in clarifying item wording, improving construct coverage, and ensuring alignment with theoretical frameworks. Third, to further validate the survey, we conducted two rounds of pilot testing with police officers from the Kentridge Police Department, allowing for iterative refinements based on respondent feedback.

To minimize the effects of common method bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003), four reverse-coded items were randomly included in the survey. Furthermore, we included a note to respondents informing them that they can cooperate with appropriate personnel to answer the questions. The final survey questionnaire items are presented in Table 1. All constructs are reflective, and each item was measured on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). Our survey items were reviewed and approved by the Institutional Review Board at the authors' university.

4.3. Data analysis

To analyze the survey data, we employed PLS-SEM using ADANCO software. PLS-SEM allows simultaneous estimation of relationships among multiple independent and latent dependent variables (Hair, Risher, Sarstedt, & Ringle, 2019), making it suitable for complex causal modelling. It allows flexibility regarding multivariate normality, accommodates small sample sizes, and supports the inclusion of both reflective and formative constructs. In addition, PLS-SEM is appropriate for exploratory analysis to test a theory under development (Lowry & Gaskin, 2014).

Our analysis proceeded in four main stages. First, we assessed and

Table 1
Constructs and Measurement Items.

Constructs and Measurement Items		Source
Recording Capability		
RC1	BWCs record conversations clearly	Semi-structured interviews
RC2	BWCs produce high quality audio	Semi-structured interviews
RC3	BWCs produce high quality video	Semi-structured interviews
RC4	BWCs record well in low light situations	Semi-structured interviews
Mobile Capability		
MC1	BWC devices are easy to use while on duty	Semi-structured interviews
MC2	BWC devices are easy to activate and deactivate while on duty	Semi-structured interviews
MC3	BWC devices remain positioned stably on the uniform while moving or running	Semi-structured interviews
MC4	BWC devices are distracting while on duty (reversed item)	Semi-structured interviews
Information Access Capability		
IAC1	Accessing footage from the storage database requires little effort	Semi-structured interviews
IAC2	The interface for accessing footage from the storage database is easy to use	Semi-structured interviews
IAC3	Accessing footage from the storage database is time consuming (reversed item)	Semi-structured interviews
IAC4	Searching and finding specific footage from the storage database is easy	Semi-structured interviews
IAC5	Retrieving specific footage from the storage database is simple	Semi-structured interviews
Information Sharing Capability		
ISHC1	Sharing footage with authorized stakeholders is difficult (reversed item)	Semi-structured interviews
ISHC2	Sharing footage with authorized stakeholders is quick	Semi-structured interviews
ISHC3	Sharing footage with authorized stakeholders is simple	Semi-structured interviews
ISHC4	The interface for sharing footage with stakeholders is easy to use	Semi-structured interviews
Information Saving Capability		
ISC1	Saving footage from BWC devices to the storage database is easy	Semi-structured interviews
ISC2	Saving footage from BWC devices to the storage database is simple	Semi-structured interviews
ISC3	Saving footage from BWC devices to the storage database is time consuming (reversed item)	Semi-structured interviews
ISC4	The interface for saving footage from BWC devices to the video storage database is intuitive	Semi-structured interviews
System Integration Capability		
SIC1	When making reports, combining BWC footage with other types of files can be done quickly	Semi-structured interviews
SIC2	When making reports, integrating BWC footage with other types of files requires just a few steps	Semi-structured interviews
SIC3	When making reports, I can easily combine BWC footage with other types of files from data storage	Semi-structured interviews
SIC4	When making reports, our officers can access all required databases at once with a single ID and password	Semi-structured interviews
Case Reporting Routine		
CR1	After having used the BWC systems, the effort required to generate case reports has been reduced	Semi-structured interviews
CR2	After having used the BWC systems, the time required to generate case reports has been reduced	Semi-structured interviews
CR3	After having used the BWC systems, officers produce more accurate case reports	Semi-structured interviews
CR4	After having used the BWC systems, officers describe events in more detail in case reports	Semi-structured interviews
CR5	After having used the BWC systems, officers frequently use BWC video footage to generate case reports	Semi-structured interviews
CR6	After having used the BWC systems, BWC footage has become an important part of generating case reports	Semi-structured interviews
Officer Supervision Routine		

Table 1 (continued)

Constructs and Measurement Items		Source
OS1	After having used the BWC systems, the effort required to supervise officers has been reduced	Semi-structured interviews
OS2	After having used the BWC systems, the time required to supervise officers has been reduced	Semi-structured interviews
OS3	After having used the BWC systems, officer supervision has become more effective	Semi-structured interviews
OS4	After having used the BWC systems, officer supervision has become easier	Semi-structured interviews
OS5	After having used the BWC systems, BWC video footage has become an important part of officer supervision	Semi-structured interviews
OS6	After having used the BWC systems, supervising officers frequently use BWC footage to review officer performance	Semi-structured interviews
Policing Performance		
PP1	After having used the BWC systems, the crime rate in our community has decreased	Smykla et al. (2016) Tankebe and Ariel (2016)
PP2	After having used the BWC systems, the number of citizen complaints has decreased	Smykla et al. (2016)
PP3	After having used the BWC systems, the number of resolved citizen complaints has increased	Tankebe and Ariel (2016)
PP4	After having used the BWC systems, the use of force by officers has decreased	Smykla et al. (2016)
Public Relationship Performance		
PRP1	After having used the BWC systems, public trust in our department has increased	Tankebe and Ariel (2016)
PRP2	After having used the BWC systems, our department's public image has improved	Tankebe and Ariel (2016)
PRP3	After having used the BWC systems, citizens have become more cooperative	Smykla et al. (2016)
PRP4	After having used the BWC systems, citizens view our officers with more respect	Smykla et al. (2016)

refined the measurement scales to ensure reliability and validity. Second, we checked for non-response bias and common method bias. Third, we estimated the structural model to evaluate the hypothesized relationships. Finally, we conducted mediation analyses to explore how changes in officer supervision and case reporting routines mediate the effects of BWC-related affordances on policing performance.

4.4. Assessment and refinement of measurement scales

Our research model (Fig. 1) included two second-order constructs: MRA (Mobile Recording Affordance) and GIT (General Information Technology Affordance). We, therefore, split the assessment of the measurement model into two stages. In the first stage, since each construct was modeled reflectively, we assessed the measurement scales to ensure internal reliability, convergence validity, and discriminant validity (Hair, Hult, Ringle, & Sarstedt, 2021). The results from the first run of PLS-SEM, which are presented in Appendix B, showed that the average variance extracted (AVE) of the case reporting routine and policing performance was 0.451 and 0.463, respectively. They are less than the threshold value of 0.5, so the convergent validity was not satisfied (Hair et al., 2021). In addition, the discriminant validity was not satisfied because the squared correlation between the case reporting routine and the officer supervision routine was 0.507, which was higher than the AVE of case reporting routine of 0.451 (Fornell & Larcker, 1981).

Thus, we refined the measurement model by examining the context of the items and removing an item PP3 ("After having used the BWC systems, the number of resolved citizen complaints has increased"), CR5 ("After having used the BWC systems, officers frequently use BWC video footage to generate case reports"), and CR6 ("After having used the BWC systems, supervising officers frequently use BWC footage to review officer performance"). The resulting measurement model, consisting of 10 constructs and 42 items, was assessed again. The results (presented in

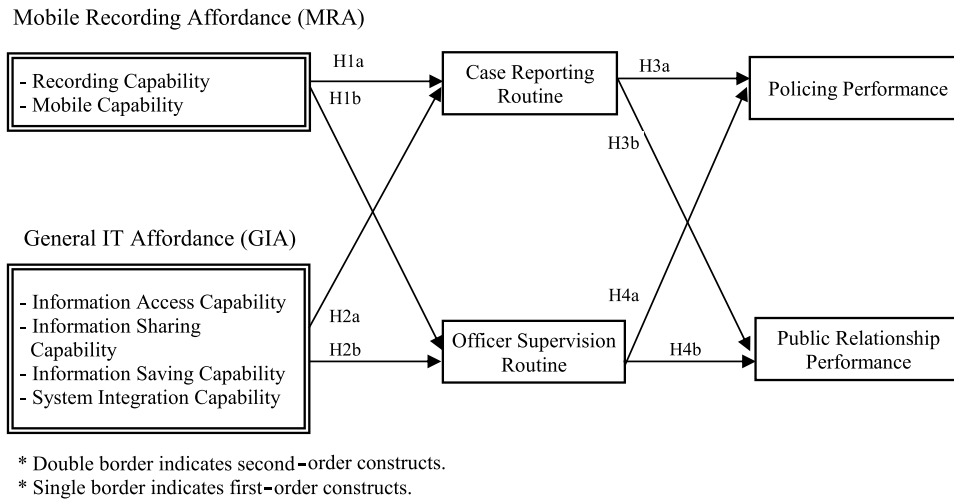


Fig. 1. Research Model.

Appendix C) showed that internal reliability, convergence validity, and discriminant validity were satisfied.

In the second stage, the second-order constructs were assessed. As mobile recording affordance and general IT affordance are second-order constructs, a disjoint two-stage approach was applied to assess the scale of the high order construct (Sarstedt, Hair Jr, Cheah, Becker, & Ringle, 2019). The results are shown in Appendix D.

4.4.1. Non-response bias and common method bias

To check for non-response bias, a *t*-test was performed to determine whether there was a difference between early and late respondents in terms of sworn officers and non-sworn officers (Armstrong & Overton, 1977). No significant differences were found. Also, to detect common method bias (Podsakoff et al., 2003), we examined four reverse-coded items as mentioned above. Ten responses were removed for missing these reverse-coded items. We performed two statistical tests were performed to further check for common method bias. First, we conducted a Harman's single factor test. The loading of the common factors was 23.06 %, indicating that common method bias was not a concern (Podsakoff et al., 2003). Second, we ensured that the correlations between any two constructs (Table B5 in Appendix C) were less than 0.9, a threshold that indicates the presence of common method bias (Pavlou, Liang, & Xue, 2007). No such correlations were detected.

4.4.2. Structural model for hypothesis testing

The structural model was estimated to test the hypotheses in the research model (Fig. 1). The results (Fig. 2) showed that the two BWC affordance variables displayed positive and significant impacts on the changes of case reporting and officer supervision routine. Specifically, MRA had positive impacts on the changes of case reporting routine ($\beta=0.205$, $p < 0.01$) and officer supervision routine ($\beta=0.195$, $p < 0.01$). Similarly, GIA had positive impacts on the changes of case reporting routine ($\beta=0.216$, $p < 0.01$) and officer supervision routine ($\beta=0.140$, $p < 0.05$). Thus, hypotheses H1(a, b) and H2(a, b) were supported.

In addition, changes in case reporting routine had a positive impact on policing performance (PP) ($\beta=0.201$, $p < 0.01$) and public relationship performance (PRP) ($\beta=0.190$, $p < 0.01$). Thus, hypotheses H3 (a, b) were supported. Finally, changes in officer supervision had a positive impact on PP ($\beta=0.300$, $p < 0.01$) and PRP ($\beta=0.393$, $p < 0.01$). So, hypotheses H4 (a, b) were also supported.

4.4.3. Mediating effect of officer supervision and case reporting routines

We performed mediation analysis to get deeper insights into how changes in officer supervision and case reporting routines mediate between the two BWC-related affordances and the two performances. For this mediation analysis, we followed the guideline of Carrión, Nitzl, and Roldán (2017) and Nitzl, Roldán, and Cepeda (2017). They emphasize the importance of examining both the indirect effect (independent

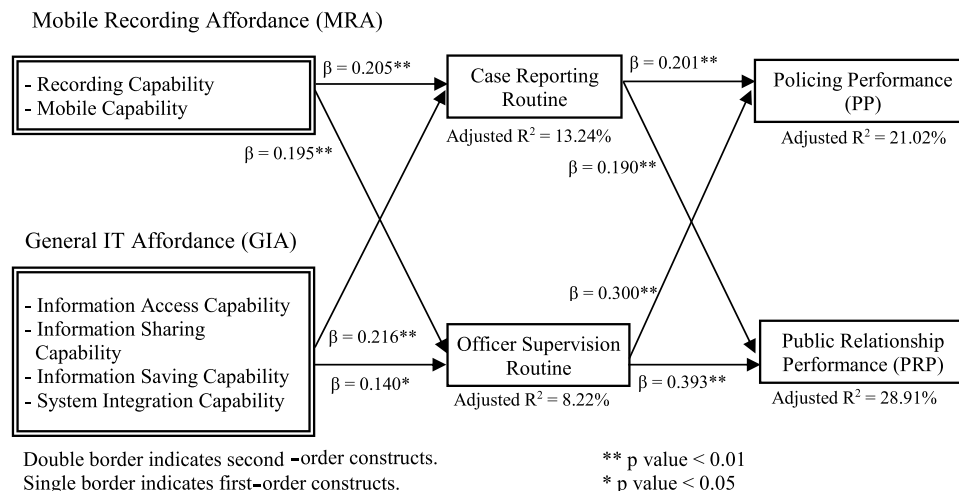


Fig. 2. Research Result.

variable → mediator → dependent variable) and the direct effect (independent variable → dependent variable) to thoroughly evaluate the presence of mediation effect.

The indirect effect, also termed the mediation effect, quantifies the extent to which the independent variable (X) influences the dependent variable (Y) through a mediator (M). The statistical significance of this effect is crucial for estimating the presence of mediation. Conversely, the direct effect represents the portion of X's influence on Y that remains after accounting for the mediator, indicating the direct relationship between X and Y.

The interplay between these effects allows us to determine the nature of mediation. A significant indirect effect coupled with a non-significant direct effect suggests full mediation, where the mediator entirely explains the relationship between X and Y. If both the direct and indirect effects are significant, it indicates partial mediation, implying that the mediator accounts for some, but not all, of the relationship. The total effect, representing the overall impact of X on Y without considering the mediator, is the sum of the direct and indirect effects (Carrion et al., 2017; Nitzl et al., 2017). All these effects are estimated and summarized in Table 2.

Our analysis first investigates whether changes in case reporting and officer supervision routines mediate the relationships between MRA and performance, specifically in terms of PP and PRP. Rows 17 through 20 in Table 2 reveal statistically significant indirect effects for the paths from MRA to both PP and PRP. This indicates that changes in case reporting and officer supervision routines do indeed mediate the impact of MRA on these two performance outcomes. Furthermore, the direct effect of MRA on PP is statistically significant (Row 8, Table 2), while the direct effect of MRA on PRP is not (Row 10, Table 2). These findings suggest that changes in case reporting and officer supervision routines partially mediate the effect of MRA on PP, as a direct relationship still exists. In contrast, these routine changes fully mediate the effect of MRA on PRP,

as the direct relationship becomes non-significant when the mediators are considered.

Second, our findings also demonstrate that changes in case reporting and officer supervision routines act as significant mediators in the relationships between GIA and performance, specifically PP and PRP. The statistically significant indirect effects for the paths from GIA to PP and PRP (as indicated in Rows 13 through 16 of Table 2) confirm that these routine changes mediate the impact of GIA on both performance outcomes. Furthermore, the direct effects of GIA on PP and PRP are statistically insignificant (Rows 2 and 4 of Table 2). This suggests that the influence of GIA on both PP and PRP is fully mediated by changes in case reporting and officer supervision routines.

The findings of the analysis reveal several intriguing narratives pertaining to the impact of technology on organizational performance. Firstly, the implementation of a new technology may bring to an organization several affordances, yet not all these affordances enhance organization-wide performance directly. Of the four direct relationships between BWC affordances (e.g., MRA and GIA) and performance in terms of PP and PRP, only one relationship - from MRA to PP - is supported by the data (as shown on Row 10 in Table 2). The remaining three relationships, including those between MRA and PRP, GIA and PP, and GIA and PRP, are not supported by the data (as shown on Rows 2, 4, and 8 in Table 2). The results suggest that if the new technology provides novel capabilities to the organization, it will likely have a direct impact on the organization's performance. MRA can be regarded as a novel capability within the police department (Cubitt et al., 2017; Lum et al., 2019) because it assists police agencies in collecting evidence that is very close to observations of police officers in the field. Therefore, it can facilitate the performance of police activities (e.g., handling citizen complaints, investigating crimes, or evaluating police behaviors), and thereby improve the entire police agency's performance. Conversely, GIA can facilitate flexibility and convenience in the process of querying, accessing, and retrieving data within the police department. However, the capabilities are not novel, and therefore its direct impact on performance may not have been fully recognized by police departments. As the results show, GIA improves performance by causing changes in case reporting and officer supervision routines, and these changed routines affect performance.

Secondly, the analysis results highlight the critical role of technology-induced routine changes in assessing the impact of technology on overall performance. The analysis results show that changes in officer supervision and case reporting routines serve as full or partial mediators between the two BWC-related affordances (e.g., MRA and GIA) and performance in terms of PP and PRP. These results stress the importance of technology-induced routine change in mediating the path effects from technology to organizational performance. In addition, the total effects of GIA and MRA on PP and PRP, as shown in Table 2, are statistically significant and higher than the direct impacts of GIA and MRA on PP and PRP. These results, therefore, imply that technology-induced changes to routine also contribute to enhance organizational performance. Consequently, the effects of BWC technology on the performance of police agencies are more pronounced when considering the routines altered by the repeated use of BWCs.

5. Discussion

Our findings offer important insights into how technology-induced routine changes can affect organizational performance. Specifically, the implementation of new cloud-based storage (i.e., GIA) for both newly generated BWC video footage and pre-existing digital evidence likely streamlined the case report writing process. This centralized storage system would have allowed for efficient collation of various evidence files under a single case number, potentially leading to quicker creation of more comprehensive and detailed case reports. Furthermore, by automatically capturing and securely storing officer-civilian interactions on video footages for later review, BWCs may have enabled

Table 2
Summary of mediation effect tests.

Row	Effect	Coefficient	Confidence Interval	
			2.5 %	97.5 %
Direct Effects				
1	GIA → Officer Supervision	0.143*	0.029	0.259
2	GIA → PRP	0.003(ns)	−0.104	0.124
3	GIA → Case Reporting	0.220*	0.097	0.347
4	GIA → PP	−0.003(ns)	−0.125	0.129
5	Officer Supervision → PRP	0.386*	0.251	0.523
6	Officer Supervision → PP	0.285*	0.160	0.427
7	MRA → Officer Supervision	0.194*	0.066	0.323
8	MRA → PRP	0.066(ns)	−0.082	0.205
9	MRA → Case Reporting	0.203*	0.071	0.343
10	MRA → PP	0.170*	0.009	0.322
11	Case Reporting → PRP	0.172*	0.033	0.315
12	Case Reporting → PP	0.158*	0.017	0.298
Indirect Effects				
13	GIA → Case Reporting → PP	0.035*	0.003	0.076
14	GIA → Case Reporting → PRP	0.038*	0.006	0.084
15	GIA → Officer Supervision → PP	0.041*	0.008	0.085
16	GIA → Officer Supervision → PRP	0.055*	0.010	0.113
17	MRA → Case Reporting → PP	0.032*	0.003	0.075
18	MRA → Case Reporting → PRP	0.035*	0.004	0.082
19	MRA → Officer Supervision → PP	0.055*	0.016	0.113
20	MRA → Officer Supervision → PRP	0.075*	0.023	0.141
Total Effects				
21	GIA → PP	0.073*	0.026	0.132
22	GIA → PRP	0.479*	0.030	0.162
23	MRA → PP	0.257*	0.110	0.397
24	MRA → PRP	0.110*	0.039	0.190

(*): significant with p-value < 0.05

(ns): not significant

police officers to focus on core policing duties and reduce the need for extensive note-taking to document their field activities. The resulting availability of audiovisual evidence could have contributed to enhancing the accuracy, completeness, and credibility of case reports, thereby potentially improving internal accountability structures and evidentiary quality to support investigative and judicial processes. These improvements in the routine of case report writing appear to have led to significant improvement in performance outcomes, including reductions in citizen complaints and use-of-force incidents. These findings and interpretation generally align with prior BWC studies that demonstrate the accountability-enhancing potential of BWCs (Ariel et al., 2017; Lum, Stoltz, Koper, & Scherer, 2019).

Equally important is the change of officer supervision routines. BWC footages accessible through GIA could have empowered supervisory officers to deliver situational and personalized performance feedback based on audiovisual evidences. These new practices likely enhanced the quality of supervisory oversight, potentially constrained excessive discretionary enforcement, and may have reinforced norms of procedural justice. Consequently, these changes in officer supervision routine could have not only standardized supervising practices in police departments but also strengthened public perceptions of fairness, transparency, and legitimacy (White & Malm, 2020). In sum, the transformation of case reporting and officer supervision routines, facilitated by BWC and GIA, appears to have led to improved organizational performance across both policing efficacy and public relations.

Notably, the results reveal distinct mediation patterns. Changes in case reporting and officer supervision routines *fully mediate* the effects of GIA on both PP and PRP, suggesting that GIA's impact on performance outcomes operates entirely through its influence on routine change. In contrast, for MRA, these routine changes *partially mediate* its effect on PP and *fully mediate* its effect on PRP. This indicates that while MRA contributes to PP both directly and indirectly through routine changes, its influence on PRP occurs solely through its effect on organizational routines. These findings underscore the critical role of routine changes in tapping the performance potential of technological affordances. They also highlight the importance of aligning technologies not just with performance goals, but with the specific organizational routines they are meant to support. Our analysis confirms that routine changes, induced by technologies, serve as key mediators between technology and performance outcomes. These results suggest that managing technologies is more an artistic feat than engineering marvel in that technological capabilities should be configured in accordance with technology use practices and organizational routines (Keltsch, Probert, & Phaal, 2011). Indeed, technology implementation is inseparable from the art of management because it frequently necessitates modifications in existing ways of working to materialize the technological capabilities for the attainment of organizational goals (Trewin, 2020).

Our findings also suggest that technology-mediated routine changes may not always involve macro-level structural overhauls that risk large-scale destabilization of organizational practices (Armbruster, Bikfalvi, Kinkel, & Lay, 2008; Layne & Lee, 2001; Tidd, 2001). Instead, our findings indicate that technology-induced micro-adjustments of routines inside organizational units can also lead to stable improvement in organizational performance at the macro-level. From this perspective, our findings demonstrate that technology can dynamically impact organizational performance in a bottom-up fashion.

5.1. Theoretical contributions

Our study differs from much of the existing BWC research, which often relies on a binary “dummy variable” approach to evaluate their impact of BWC on policing performance (e.g., Use of BWCs → Policing Outcomes) (Alda, 2023; Ariel, Farrar, & Sutherland, 2015; Douglas, 2021; Pyo, 2021; White, Gaub, & Todak, 2018). These studies typically overlook the underlying structural mechanisms through which BWC technology can affect organizational performance. In contrast, we

developed an integrated affordance theoretic model, and applied it to estimate not only the direct technological effects but also the structural mechanisms, specifically through MRA and GIA, that drive changes in organizational routine and performance. To our knowledge, our study, which leverages national survey from US police department, is among the first to apply the affordance perspective to examine the structural mechanisms by which BWCs influences organizational routines and performance (Choi et al., 2023; Gaub & White, 2020; Lum et al., 2020; Lum et al., 2019).

Beyond that, several theoretical contributions follow from our research. While previous qualitative research has focused on explicating technology-induced changes in organizational routines, it has overlooked whether these changes subsequently affect organizational performance (Boillat et al., 2015; Goh et al., 2011; Hanelt et al., 2017; Krancher et al., 2018; Leidner et al., 2018; Leonardi, 2011). In contrast, although prior quantitative studies have examined the direct effects of technological affordances on organizational performance, they have typically not considered the mediating role of technology-induced changes in organizational routines (Benitez, Braojos, Pavlou, & Llorens, 2023; Chatterjee et al., 2015; 2020; Chen, Wei, Davison, & Rice, 2020; Duan et al., 2024; Majchrzak et al., 2013; Pee, 2018; Wang et al., 2020; Zhu & Jin, 2024). We address this gap between qualitative and quantitative approaches by presenting and empirically testing our integrated affordance theoretic model. In particular, we show that technological affordances (i.e., such as MRA and GIA) reshape core organizational routines (i.e., case reporting and officer supervision), which, in turn, affect organizational performance (i.e., PP and PRP).

Also, our study adds contributions to strategic IT management research by empirically validating a materialist perspective on organizational development. While strategic IT management research has long argued that IT investment increases business process efficiency

(Hitt & Brynjolfsson, 1997; Kohli & Grover, 2008; Mithas & Rust, 2016; Schryen, 2013), it has largely overlooked the material capabilities of technologies used for organizational tasks. This implies that much of the assumed IT business value relies on proxy variables (e.g., IT budgets, expenditures, or labor assets to represent technology), presuming that higher IT investment is likely to lead to better IT performance (Ewens, Peters, & Wang, 2025; Mithas & Rust, 2016; Sabherwal, Sabherwal, Havakhor, & Steelman, 2019; Saunders & Brynjolfsson, 2016). Our study demonstrates, however, that technology, whether hardware or software, can add value when its specific material capabilities improve existing work practices to achieve organizational goals. In this regard, our integrated affordance theory model offers a new lens for understanding the IT–organizational performance link by emphasizing the performative and material dimensions of technology. To our knowledge, this is the first study to present and empirically validate such a model.

5.2. Implications for management

This study offers valuable practical insights for organizations and their leadership when adopting new technologies. Leaders should not only plan for intended uses but also remain open to unexpected uses, and emergent benefits that arise from the use of diverse technological affordances. Importantly, they should be mindful that not all technological affordances directly enhance organizational performance. Instead, they often reshape underlying organizational routines to improve organizational performance. For example, police agencies that restructured their existing IT infrastructure to leverage the integrative capacity of BWC cloud storage systems reported unanticipated enhancements in case reporting and officer supervision routines (Guzik et al., 2021). To maximize the value of technological affordances, leaders are encouraged to foster a culture of experimentation and flexibility in routine design. This includes encouraging frontline staff to explore new ways of using technology, monitoring emergent practices to achieve their goals, and institutionalizing beneficial deviations through formal routine updates. Additionally, leaders should invest in training

and cross-functional collaboration to ensure that the full range of a technological affordances is recognized and effectively integrated into daily operations.

Leadership should recognize that technological capabilities are relational and create composite capabilities which cannot be reduced solely to users or technologies. Therefore, once new technologies are implemented, leadership needs to evaluate if emerging organizational practices resulting from the introduction of new technologies are well-aligned with organizational goals. For example, the easy recording and accessibility features of BWCs were reported to support police officers in writing case reports (Guzik et al., 2021). Thus, leaders should assess whether the interaction between technological affordances and user practices fosters positive compound capabilities that align with organizational objectives.

Lastly, building on the relational nature of technological affordances, we suggest that IT project managers carefully assess how new technological capabilities complement existing ones. For example, officers considered certain aspects of BWC technology as a constraint, due to lower resolution for capturing evidence than higher-resolution digital still camera (Guzik et al., 2021; Miranda, 2022). As a result, some detectives still carried dedicated digital cameras, and expressed concerns about officers using BWCs for crime scene documentation.

5.3. Limitations and future research

Our study has limitations that future research should address. First, affordance theory maintains that technology possesses agential capabilities to empower or constrain organizational users. Our research primarily focused on the “empowerment” side of BWC systems. Future research should investigate its “constraint” side to build a holistic understanding of technological affordance. For instance, increasing reliance on video footage led some officers to omit detailed textual descriptions, opting instead for simple references to specific video segments, thereby diminishing report quality (Boivin & Gendron, 2021). Given that textual documentation of focal events with cogent interpretation is of utmost importance in court proceedings, this raises a concern that heavy reliance on video footage for evidence documentation may erode officers’ skills to interpret and articulate specific cases in written documents. This highlights the risk of negative compound capabilities that arise from over-dependence on new technology for tasks requiring active and interpretative engagement of human intelligence with written texts. Thus, employing our integrated affordance theoretic model as a conceptual framework, qualitative researchers could use grounded theory or ethnographic research methods involving in-depth interviews and/or close observation of technology use to examine how certain human skills might be diminished or lost when human intelligent tasks are heavily reliant on new technological capabilities. In addition, future research can examine what strategies organizations can adopt to retain these human intelligent skills and how organizations can design new practices to develop positive composite capabilities that can maximize both human and technological capabilities.

Second, the survey data is self-reported, and raises the potential for social desirability bias, which is prevalent in survey-based research (Guedes, 2017). This is a particular concern with our dependent variables (i.e., policing performance and public relationship performance), where agency representatives might feel pressured to report positive outcomes. Objective measures of policing performance are available (e.

g., the FBI’s Uniform Crime Report² or the Bureau of Justice Statistics’ National Crime Victimization Survey³). While self-reports proved more feasible for matching performance outcomes to reported technology-induced routine changes in our study, future studies could integrate these data sources to provide a more holistic picture of technology-induced routine changes and their effects on organizational performance.

Third, our findings are based on data collected from U.S. police agencies, which limits their generalizability. The structure, policies, and technological adoption of law enforcement agencies vary across different countries, which may influence how BWCs shape organizational routines. Additionally, cultural, legal, and institutional differences may affect how officers interact with and utilize BWCs, leading to distinct patterns of potential routine change. Also, our study focuses on a specific set of affordances and routines, which may not capture the full range of effects that BWCs have in other policing environments. Future research should extend this analysis by incorporating data from diverse cultures, countries, and alternative technologies to assess the broader applicability of our integrated affordance theoretic model. For example, comparative case studies between law enforcement agencies in countries with different legal systems (e.g., adversarial vs. inquisitorial) could reveal how institutional contexts shape the enactment of BWC affordances. Also, longitudinal studies could also be valuable in examining how affordances evolve over time as both technology and user expertise mature.

6. Conclusion

Building on the qualitative and quantitative literature on affordances and routine change, we developed an integrated affordance theory model to illustrate the pathway through which technological affordances influence organizational performance. We applied this framework to the case of BWC implementation in U.S. police agencies with survey data from 338 agencies. Using PLS-SEM, our analysis confirm that technology-induced routine change plays a central role in mediating the relationship between technological affordances and organizational performance. Our integrated model offers a new perspective on IT and organizational performance by emphasizing that technologies possess material capabilities that can drive the changes of organizational routines, ultimately affecting performance outcomes. Thus, our integrated model and empirical findings contribute to the literature on affordance theory, routine theory, and the IT -organizational performance.

Funding

This research was supported by the U.S. National Science Foundation under Grant No. 1734632.

CRediT authorship contribution statement

Onook Oh: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Ronald Ramirez:** Writing – review & editing, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Dung Tien Nguyen:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Keith Guzik:** Writing – review & editing, Methodology, Investigation,

² Please refer to the following link for detailed information on the FBI’s Uniform Crime Report:

<https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr>

³ Detailed information regarding the Bureau of Justice Statistics’ National Crime Victimization Survey can be accessed via the following link: <https://bjs.ojp.gov/programs/ncvs>

Conceptualization. **Tony W. Tong:** Writing – review & editing, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper. In addition, the findings, interpretations, and conclusions expressed in this article are those of the authors and do not necessarily reflect the views of the National Science Foundation. The authors declare no conflicts of interest.

Appendix A. Description of the Semi-structure Interviews

We conducted semi-structured interviews with 10 police employees at the Kentridge Police Department, a higher-income city suburb of approximately 50,000 people. The department had approximately 115 police employees and 65 BWCs. It received national recognition for its use of body-worn cameras (BWCs). The objective of the interviews was to gather input from the department, identify any gaps in our research model and constructs, and gain an initial understanding of how the department used and experienced BWCs. The sample included officers from diversified ranks. The 10 police employees who were interviewed included one patrol commander, two detectives, one IT specialist, and six staff members with other duties (e.g., public relations, media, evidence, records, and support services). The interview protocol included topics drawn from existing BWC literature (White et al., 2018) as well as others reflective of affordance theory: *How do the cameras work? How are videos labelled, uploaded, stored, and accessed from the cameras? Is the video software system a stand-alone system or does it communicate with other (cloud-based) management systems? What has been your experience with BWCs (both positive and negative)? How have BWCs changed the way you think about your work or policing in general?* The sessions were audio recorded and later transcribed by a professional transcription service.

Table A1
Description of interviewees

Name	Department	Position	Race	Gender
Specialist Jenkins	Kentridge	IT Specialist	White	Male
Specialist Silkes	Kentridge	Public Relations	White	Male
Technician Rooks	Kentridge	Media Duplication	White	Female
Manager Toland	Kentridge	Evidence Manager	White	Female
Manager Knox	Kentridge	Records Manager	White	Female
Officer Leonard	Kentridge	Internal Affairs	White	Female
Detective Gates	Kentridge	Detective	White	Male
Detective Hawthorn	Kentridge	Detective	White	Male
Commander Lenox	Kentridge	Patrol Commander	White	Male
Manager Hand	Kentridge	Support Services Manager	White	Male

Reference

White, M. D., Todak, N., & Gaub, J. E. (2018). Examining body-worn camera integration and acceptance among police officers, citizens, and external stakeholders. *Criminology & public policy*, 17(3), 649–677.

Appendix B. Results from the first run PLS

Table B1
Constructs, measurement items and loadings

Recording Capability		
RC1	BWCs record conversations clearly	0.874
RC2	BWCs produce high quality audio	0.925
RC3	BWCs produce high quality video	0.873
RC4	BWCs record well in low light situations	0.749
Mobile Capability		
MC1	BWC devices are easy to use while on duty	0.877
MC2	BWC devices are easy to activate and deactivate while on duty	0.840
MC3	BWC devices remain positioned stably on the uniform while moving or running	0.641
MC4	BWC devices are distracting while on duty (reversed item)	0.554
Information Access Capability		
IAC1	Accessing footage from the storage database requires little effort	0.775
IAC2	The interface for accessing footage from the storage database is easy to use	0.851
IAC3	Accessing footage from the storage database is time consuming (reversed item)	0.676
IAC4	Searching and finding specific footage from the storage database is easy	0.804
IAC5	Retrieving specific footage from the storage database is simple	0.883
Information Sharing Capability		
ISHC1	Sharing footage with authorized stakeholders is difficult (reversed item)	0.752
ISHC2	Sharing footage with authorized stakeholders is quick	0.924
ISHC3	Sharing footage with authorized stakeholders is simple	0.942
ISHC4	The interface for sharing footage with stakeholders is easy to use	0.908
Information Saving Capability		
ISC1	Saving footage from BWC devices to the storage database is easy	0.915
ISC2	Saving footage from BWC devices to the storage database is simple	0.923
ISC3	Saving footage from BWC devices to the storage database is time consuming (reversed item)	0.665

(continued on next page)

Table B1 (continued)

Recording Capability		
ISC4	The interface for saving footage from BWC devices to the video storage database is intuitive	0.658
System Integration Capability		
SIC1	When making reports, combining BWC footage with other types of files can be done quickly	0.910
SIC2	When making reports, integrating BWC footage with other types of files requires just a few steps	0.905
SIC3	When making reports, I can easily combine BWC footage with other types of files from data storage	0.934
SIC4	When making reports, our officers can access all required databases at once with a single ID and password	0.555
Case Reporting Routine		
CR1	After having used the BWC systems, the effort required to generate case reports has been reduced	0.733
CR2	After having used the BWC systems, the time required to generate case reports has been reduced	0.543
CR3	After having used the BWC systems, officers produce more accurate case reports	0.812
CR4	After having used the BWC systems, officers describe events in more detail in case reports	0.767
CR5	After having used the BWC systems, officers frequently use BWC video footage to generate case reports	0.592
CR6	After having used the BWC systems, BWC footage has become an important part of generating case reports	0.524
Officer Supervision		
OS1	After having used the BWC systems, the effort required to supervise officers has been reduced	0.702
OS2	After having used the BWC systems, the time required to supervise officers has been reduced	0.659
OS3	After having used the BWC systems, officer supervision has become more effective	0.807
OS4	After having used the BWC systems, officer supervision has become easier	0.832
OS5	After having used the BWC systems, BWC video footage has become an important part of officer supervision	0.706
OS6	After having used the BWC systems, supervising officers frequently use BWC footage to review officer performance	0.583
Policing Performance		
PP1	After having used the BWC systems, the crime rate in our community has decreased	0.642
PP2	After having used the BWC systems, the number of citizen complaints has decreased	0.744
PP3	After having used the BWC systems, the number of resolved citizen complaints has increased	0.618
PP4	After having used the BWC systems, the use of force by officers has decreased	0.710
Public Relationship Performance		
PRP1	After having used the BWC systems, public trust in our department has increased	0.798
PRP2	After having used the BWC systems, our department's public image has improved	0.828
PRP3	After having used the BWC systems, citizens have become more cooperative	0.794
PRP4	After having used the BWC systems, citizens view our officers with more respect	0.827

Table B2
Construct Reliability

Constructs	Cronbach's alpha (α)	Composite Reliability	Average variance extracted (AVE)
Recording Capability	0.878	0.917	0.735
Mobile Capability	0.708	0.824	0.548
Information Access Capability	0.860	0.899	0.641
Information Sharing Capability	0.906	0.935	0.783
Information Saving Capability	0.806	0.874	0.641
System Integration Capability	0.848	0.903	0.707
Case Reporting Routine	0.757	0.827	0.451*
Officer Supervision Routine	0.811	0.864	0.518
Policing Performance	0.610	0.774	0.463*
Public Relationship Performance	0.828	0.886	0.659
(*) Violate convergent validity			

Table B3
Discriminant Validity: Heterotrait-Monotrait Inference

Constructs	RC	MC	IAC	IShC	ISC	SIC	CR	OS	PP	PRP
Recording Capability (RC)										
Mobile Capability (MC)	0.876									
Information Access Capability (IAC)	0.613	0.777								
Information Sharing Capability (IShC)	0.408	0.599	0.682							
Information Saving Capability (ISC)	0.562	0.704	0.744	0.531						
System Integration Capability (SIC)	0.414	0.560	0.606	0.683	0.492					
Case Reporting Routine (CR)	0.451	0.496	0.422	0.301	0.368	0.472				
Officer Supervision Routine (OS)	0.373	0.449	0.340	0.233	0.277	0.389	0.950			
Policing Performance (PP)	0.486	0.641	0.415	0.268	0.373	0.443	0.719	0.790		
Public Relationship Performance (PRP)	0.304	0.403	0.242	0.204	0.170	0.383	0.626	0.711	0.926	

95 % bootstrap quantiles

Table B4
Discriminant Validity: Fornell-Larcker Criterion

Constructs	RC	MC	IAC	IShC	ISC	SIC	CR	OS	PP	PRP
Recording Capability (RC)	0.735									
Mobile Capability (MC)	0.416	0.548								
Information Access Capability (IAC)	0.186	0.286	0.641							
Information Sharing Capability (IShC)	0.070	0.152	0.280	0.783						
Information Saving Capability (ISC)	0.161	0.231	0.298	0.133	0.641					
System Integration Capability (SIC)	0.077	0.128	0.212	0.302	0.109	0.707				
Case Reporting Routine (CR)	0.087	0.088	0.079	0.032	0.048	0.104	0.451			
Officer Supervision Routine (OS)	0.054	0.069	0.046	0.013	0.021	0.060	0.507*	0.518		
Policing Performance (PP)	0.072	0.118	0.049	0.012	0.031	0.055	0.184	0.242	0.463	
Public Relationship Performance (PRP)	0.031	0.057	0.018	0.007	0.002	0.064	0.197	0.274	0.353	0.659

AVEs in the diagonal and bold font

Squared correlations between two constructs below the diagonal.

Appendix C. Assessment of First-Order Constructs

To assess internal reliability, the composite reliability (CR), Cronbach's alpha, and average variance extracted (AVE) of each construct were calculated. Conventionally, the CR above 0.7, Cronbach's alpha above 0.7, and AVE above 0.5 are recommended for the internal reliability (Hair et al., 2021). As shown in Table C2, except policing performance, other constructs have satisfied the thresholds of CR, Cronbach's alpha, and AVE, confirming their reliability. Policing performance construct has Cronbach's alphas at 0.623, lower than recommended, but still above acceptable limit of 0.6 (Peterson, 1994). Hence, its reliability can be acceptable. In addition, the average variance extracted (AVE) of all constructs is higher than 0.5, showing that convergent validity is satisfied. To assess discriminant validity, Heterotrait-Monotrait (HTML) and Fornell & Larcker criteria were used (Henseler, Ringle, & Sarstedt, 2015). HTML inferences are lower than 1, and the correlation coefficients between pairs of constructs were well below the square root of AVEs (Tables C3 and C4), ensuring satisfactory discriminant validity.

Table C1
Constructs, measurement items and loadings

Recording Capability		
RC1	BWCs record conversations clearly	0.876
RC2	BWCs produce high quality audio	0.925
RC3	BWCs produce high quality video	0.869
RC4	BWCs record well in low light situations	0.751
Mobile Capability		
MC1	BWC devices are easy to use while on duty	0.877
MC2	BWC devices are easy to activate and deactivate while on duty	0.838
MC3	BWC devices remain positioned stably on the uniform while moving or running	0.645
MC4	BWC devices are distracting while on duty (reversed item)	0.552
Information Access Capability		
IAC1	Accessing footage from the storage database requires little effort	0.774
IAC2	The interface for accessing footage from the storage database is easy to use	0.850
IAC3	Accessing footage from the storage database is time consuming (reversed item)	0.677
IAC4	Searching and finding specific footage from the storage database is easy	0.802
IAC5	Retrieving specific footage from the storage database is simple	0.884
Information Sharing Capability		
ISHC1	Sharing footage with authorized stakeholders is difficult (reversed item)	0.750
ISHC2	Sharing footage with authorized stakeholders is quick	0.926
ISHC3	Sharing footage with authorized stakeholders is simple	0.944
ISHC4	The interface for sharing footage with stakeholders is easy to use	0.907
Information Saving Capability		
ISC1	Saving footage from BWC devices to the storage database is easy	0.904
ISC2	Saving footage from BWC devices to the storage database is simple	0.915
ISC3	Saving footage from BWC devices to the storage database is time consuming (reversed item)	0.665
ISC4	The interface for saving footage from BWC devices to the video storage database is intuitive	0.681
System Integration Capability		
SIC1	When making reports, combining BWC footage with other types of files can be done quickly	0.910
SIC2	When making reports, integrating BWC footage with other types of files requires just a few steps	0.905
SIC3	When making reports, I can easily combine BWC footage with other types of files from data storage	0.934
SIC4	When making reports, our officers can access all required databases at once with a single ID and password	0.554
Case Reporting Routine		
CR1	After having used the BWC systems, the effort required to generate case reports has been reduced	0.785
CR2	After having used the BWC systems, the time required to generate case reports has been reduced	0.566
CR3	After having used the BWC systems, officers produce more accurate case reports	0.819
CR4	After having used the BWC systems, officers describe events in more detail in case reports	0.784
CR5	After having used the BWC systems, officers frequently use BWC video footage to generate case reports	removed
CR6	After having used the BWC systems, BWC footage has become an important part of generating case reports	removed
Officer Supervision		
OS1	After having used the BWC systems, the effort required to supervise officers has been reduced	0.712
OS2	After having used the BWC systems, the time required to supervise officers has been reduced	0.663
OS3	After having used the BWC systems, officer supervision has become more effective	0.803
OS4	After having used the BWC systems, officer supervision has become easier	0.837

(continued on next page)

Table C1 (continued)

Recording Capability		
OS5	After having used the BWC systems, BWC video footage has become an important part of officer supervision	0.697
OS6	After having used the BWC systems, supervising officers frequently use BWC footage to review officer performance	0.573
Policing Performance		
PP1	After having used the BWC systems, the crime rate in our community has decreased	0.709
PP2	After having used the BWC systems, the number of citizen complaints has decreased	0.761
PP3	After having used the BWC systems, the number of resolved citizen complaints has increased	removed
PP4	After having used the BWC systems, the use of force by officers has decreased	0.796
Public Relationship Performance		
PRP1	After having used the BWC systems, public trust in our department has increased	0.793
PRP2	After having used the BWC systems, our department's public image has improved	0.823
PRP3	After having used the BWC systems, citizens have become more cooperative	0.799
PRP4	After having used the BWC systems, citizens view our officers with more respect	0.831

Table C2

Construct Reliability

Constructs	Cronbach's alpha (α)	Composite Reliability	Average variance extracted (AVE)
Recording Capability	0.878	0.917	0.735
Mobile Capability	0.708	0.824	0.548
Information Access Capability	0.860	0.899	0.641
Information Sharing Capability	0.906	0.935	0.783
Information Saving Capability	0.806	0.874	0.640
System Integration Capability	0.848	0.903	0.707
Case Reporting Routine	0.729	0.831	0.556
Officer Supervision Routine	0.811	0.864	0.518
Policing Performance	0.623	0.800	0.572
Public Relationship Performance	0.828	0.885	0.659

Table C3

Discriminant Validity: Heterotrait-Monotrait Inference

Constructs	RC	MC	IAC	IShC	ISC	SIC	PP	PRP	CR	OS
Recording Capability (RC)										
Mobile Capability (MC)	0.876									
Information Access (IAC)	0.613	0.777								
Information Sharing (IShC)	0.408	0.599	0.682							
Information Saving (ISC)	0.562	0.704	0.744	0.531						
System Integration (SIC)	0.414	0.560	0.606	0.683	0.492					
Policing Performance (PP)	0.476	0.507	0.416	0.306	0.358	0.498				
Public Relationship Performance (PRP)	0.373	0.449	0.340	0.233	0.277	0.389	0.958			
Case Reporting (CR)	0.444	0.584	0.375	0.248	0.278	0.412	0.731	0.714		
Officer Supervision (OS)	0.304	0.403	0.242	0.204	0.170	0.383	0.686	0.711	0.896	

95 % bootstrap quantiles

Table C4

Discriminant Validity: Fornell-Larcker Criterion

Constructs	RC	MC	IAC	IShC	ISC	SIC	PP	PRP	CR	OS
Recording Capability (RC)	0.735									
Mobile Capability (MC)	0.415	0.548								
Information Access (IAC)	0.185	0.286	0.641							
Information Sharing (IShC)	0.070	0.152	0.279	0.783						
Information Saving (ISC)	0.157	0.228	0.300	0.132	0.640					
System Integration (SIC)	0.077	0.128	0.212	0.303	0.108	0.707				
Policing Performance (PP)	0.056	0.089	0.035	0.010	0.012	0.046	0.572			
Public Relationship Performance (PRP)	0.031	0.057	0.018	0.007	0.002	0.065	0.342	0.659		
Case Reporting (CR)	0.092	0.086	0.073	0.032	0.042	0.109	0.167	0.212	0.556	
Officer Supervision (OS)	0.054	0.069	0.046	0.013	0.021	0.061	0.193	0.274	0.477	0.518

AVEs in the diagonal and bold font

Squared correlations between two constructs below the diagonal.

Table C5
Inter-Construct Correlations

Constructs	RC	MC	IAC	IShC	ISC	SIC	PP	PRP	CR	OS
Recording Capability (RC)	1.000									
Mobile Capability (MC)	0.644	1.000								
Information Access (IAC)	0.430	0.535	1.000							
Information Sharing (IShC)	0.264	0.390	0.528	1.000						
Information Saving (ISC)	0.397	0.478	0.548	0.364	1.000					
System Integration (SIC)	0.278	0.358	0.460	0.550	0.328	1.000				
Policing Performance (PP)	0.236	0.299	0.188	0.099	0.108	0.215	1.000			
Public Relationship Performance (PRP)	0.176	0.238	0.134	0.081	0.044	0.254	0.585	1.000		
Case Reporting (CR)	0.303	0.293	0.269	0.180	0.206	0.330	0.409	0.461	1.000	
Officer Supervision (OS)	0.233	0.262	0.215	0.116	0.143	0.247	0.440	0.524	0.690	1.000

Reference

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Appendix D. Assessment of Second-Order Constructs

As mobile recording affordance is a second-order formative construct, the indicator weights should be significant, and multicollinearity among the indicators should not exist. Table D1 demonstrates that the weights of the formative indicators are statistically significant (T-statistic value >1.96, $p < 0.05$). Also, the variance inflation factor (VIF) statistics are below the threshold value of 3, indicating the absence of multicollinearity (Hair et al., 2021). Because general IT affordance is a second-order reflective construct, we assessed its internal reliability, convergence validity, and discriminant validity. As shown in Table D2, composite reliability (CR) and Cronbach's alpha (α) are higher than 0.7, and AVE is higher than 0.5, indicating that internal consistency is satisfactory. The AVE is higher than 0.5, showing that convergence validity is satisfied. Also, discriminant validity is satisfied, because HTML inferences are lower than 1 (in Table D3), and the correlation coefficients between pairs of constructs were well below the square roots of their respective AVEs (Table D4) (Henseler et al., 2015).

Table D1
Assessment of Mobile recording affordance Construct

Constructs	Weight (t-value)	VIF
Mobile recording affordance		
Record Capability	0.544 (11.047)	1.71
Mobile Capability	0.559 (11.953)	1.71

Table D2
Assessment of general IT affordance Construct

Construct	Loading
General IT affordance ($\alpha = 0.775$, CR = 0.853, AVE = 0.593)	
Information Access	0.825
Information Sharing	0.749
Information Saving	0.686
System Integration	0.811

Table D3
Discriminant Validity: Heterotrait-Monotrait Inference

Construct	GIA	PP	PRP	CR	OS
General IT Affordance (GIA)					
Policing Performance (PP)	0.410				
Public Relationship Performance (PRP)	0.306	0.896			
Case Report (CR)	0.499	0.731	0.686		
Officer Supervision (OS)	0.385	0.714	0.711	0.958	

95 % bootstrap quantiles

Table D4
Discriminant Validity: Fornell-Larcker Criterion

Construct	GIA	PP	PRP	CR	OS
General IT Affordance (GIA)	0.593				
Policing Performance (PP)	0.061	0.518			
Public Relationship Performance (PRP)	0.035	0.275	0.660		
Case Report (CR)	0.108	0.482	0.214	0.558	
Officer Supervision (OS)	0.045	0.194	0.344	0.168	0.572

AVEs in the diagonal and bold font

Squared correlations between two constructs below the diagonal.

Reference

Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2021). *A primer on partial least squares structural equation modeling (PLS-SEM)*. Sage publications.
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